COMMERCIAL ITEM DESCRIPTION

BLOWPIPE, SOLDERING, DENTAL LABORATORY

The General Services Administration has authorized the use of this Commercial Item Description in preference to Interim Federal Specification GG-B-00501C (DSA-DM) dated 19 June 1975.

This commercial item description covers a gas-air operated blowpipe, suitable for melting precious metals in dental laboratory procedures.

Salient characteristics:

Materials.

Bronze. The gas and air mixing chamber shall be cast bronze conforming to Copper Alloy No. 4A of ASTM B30.

Brass. The flame tube shall be fabricated of seamless tubing conforming to Copper Alloy No. 330 of ASTM B135, and shall have a working pressure suitable for the purpose. All other brass components shall conform to Copper Alloy numbers 268 of ASTM B36 or 360 of ASTM B16.

Design and dimensions. The blowpipe shall conform to figure 1 for design and dimensions. The blowpipe shall be furnished complete with gas and air tube connectors, two control valves, mixing chamber, flame tube with regulating nut attached, and three flame tube tips. The blowpipe shall be dual control with the gas and air individually controlled. All parts of the blowpipe shall be brass, as specified herein, except the gas and air mixing chamber shall be cast bronze, as specified herein.

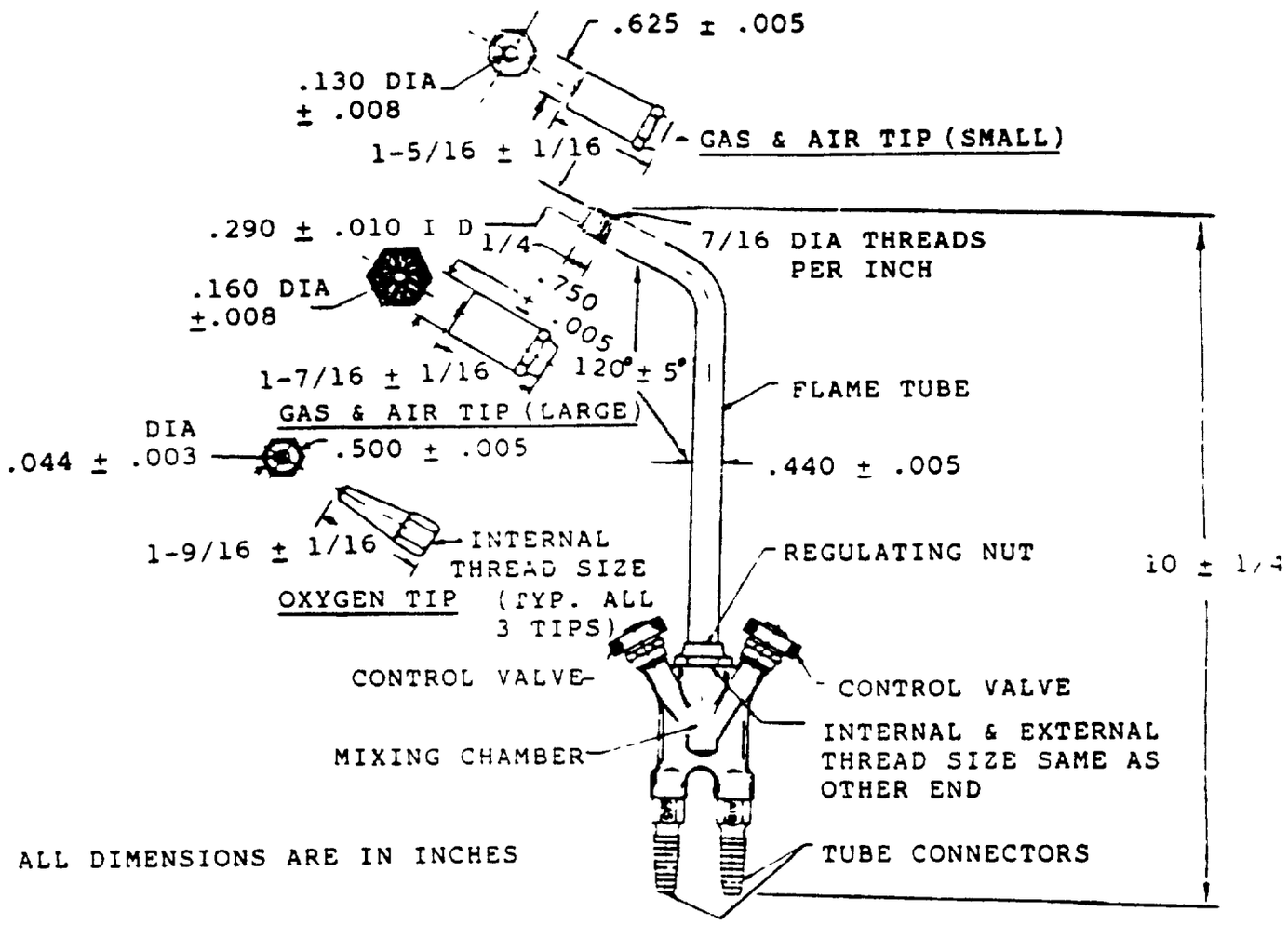
Control valves. The blowpipe shall be equipped with two sensitive needle control valves located opposite each other on the mixing chamber for easy control by thumb and forefinger, and for accurate adjustment of the flame size and intensity with each tip.

Mixing chamber. The gas and air mixing chamber shall be so designed that air passing through the chamber shall create a vacuum on the gas inlet. The mixing chamber shall act as a handgrip.

AMSC N/A FSC 6520

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BLOWPIPE

FIGURE 1

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Gas and air tube connectors. The free ends of the gas and air tubes shall be designed for attachment of rubber tubing having an inside diameter of 1/4 inch. Each tube connector shall be permanently marked for "GAS" or "AIR" as applicable.

Tips. Three tips shall be furnished with each unit. The tips shall be so constructed as to produce a dark purplish blue outer flame, a light blue center cone, and an inner cone of darker blue, when the gas and air volumes are properly proportioned. Tips shall be threaded for attachment to the flame tube, and shall be detachable with the aid of a wrench. One large and one small gas and air tip shall be furnished. The tips shall be designed to produce a full range of flame characteristics from a fine needle-point with the small tip for fine soldering, to a full brush flame with the large tip for melting precious metals. One oxygen tip shall be furnished. The tip shall be designed for operation with oxygen on types of gases specified herein, to develop a higher temperature flame than that which can be obtained with gas and compressed air.

Flame tube and regulating nut. The flame tube shall be easily adjustable without the aid of a wrench, so that the flame can be directed in a circular range. The tube and the mixing chamber shall form a leakproof union and the tube shall be locked in the desired position by a screw-type hexagonal regulating nut.

Performance. The blowpipe shall be suitable for soldering and melting precious metals, when tested as specified herein. The blowpipe shall operate efficiently on artificial gas, natural gas, mixtures of artificial and natural gases, propane, butane and acetylene when mixed with compressed air or oxygen, and shall be adaptable for such use by interchanging the tips. The mixing chamber shall be so designed as to provide a well mixed volume of gas and air to produce an intense steady flame perfectly formed. The blowpipe shall not sputter or blow out at any working air pressure between 3 and 30 pounds. At 15 pounds air pressure, it shall be possible to vary the flame from oxidizing to reducing by suitable adjustment of the control valves.

Finish. The mixing chamber shall have a crinkle finish of baked black enamel. All other exposed surfaces shall be nickel plated.

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Identification marking. The blowpipe shall be permanently and legibly stamped with the name or registered trademark of the manufacturer. Each unit shall be furnished with adequate and accurate instructions for use and maintenance of the blowpipe.

Workmanship. The blowpipe shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One blowpipe, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.5 (percent defective) and an inspection level of S-2. The unit of product for examination purposes shall be one blowpipe. Examination shall include, but shall not be limited to the defects listed below.

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Defects.

Design of blowpipe not as specified.
Components of blowpipe missing.
Threads are stripped, or bind.
Identification marking not present, not permanent, illegible, or incomplete.
Plating of components show signs of peeling or flaking.
Component parts not free from cracks, burrs, and rough or sharp edges.
Dimensions not as specified.

Sampling for tests. Sampling for tests, shall be conducted in accordance with MIL-STD-105, and as specified below, except that the acceptance number shall be zero for all sample sizes. The unit of product shall be as indicated below.

Test criteria.

Characteristic	Unit of product	Inspection level
Chemical Analysis	Not applicable	*
Performance	Blowpipe	S-2

*A sufficient quantity of each lot of raw material used in the fabrication of the components of the blowpipe shall be selected for chemical analysis.

Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

Tests. Tests shall be conducted to determine compliance with the requirements of this specification.

Chemical analysis. In the absence of certificates of quality, chemical analysis of the individual components of the blowpipe shall be determined in accordance with Method 111 of FED-STD-151.

Performance test. Before testing the blowpipe, all connections in the test set-up shall be checked for tightness and gas leaks by the use of soapsuds. All leaking connections shall be corrected before beginning the tests. The blowpipe tips, upon manipulation of the control valves, shall produce the flames specified herein.

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Metric products: Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

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NOTE: The following National Stock Number is covered by this document:

National Stock Number	Item Identification
6520-00-502-0000	BLOWPIPE, SOLDERING, DENTAL, LABORATORY

Ordering data. Intermediate/exterior package quantities and labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

PROJECT NO. 6520-2224

LOCATION: NCR Disk No. 7

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53846	BLOWPIPE, SOLDERING, DENTAL LABORATORY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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